

Human Vision vs. Computer Vision: A Readability Study in a Virtual Reality Environment

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ABSTRACT

Text plays an important role in conveying information to users in a virtual reality (VR) environment. Both VR software and hardware are evolving rapidly to improve text display quality. However, evaluation of text readability still relies on human participants. In this study, cloud computer vision was used to evaluate text readability in VR. Human subjects were recruited to test the same text scenarios. The cloud computer vision-based approach produced results that were consistent with human vision-based recognition. The use of computer vision to automate text readability evaluation could significantly reduce the overall effort and time in developing readable text in VR.

Keywords: Text readability, Evaluation methods, Computer vision, Transportation, Road sign readability, User study, Human factors.

Index Terms: Human-centered computing – Visualization - Visualization design and evaluation methods; Applied computing - Operations research – Transportation

1 INTRODUCTION

While the most obvious advantage of VR lies in the immersive experiences through a multimodal presentation, texts (e.g., UI texts and text-based signs) are still prominent ways of conveying information to the users [1,2]. Researchers and engineers have investigated hardware and software factors influencing the readability of texts (e.g., head-mounted displays, rendering pipelines, and shaders) to increase text display qualities [3–6]. While the hardware and software technologies are evolving rapidly, the text readability evaluation method has not changed and still predominantly relies on feedback from human users. User studies are costly and require time and effort on both - users and developers [7,8]. The COVID-19 pandemic further exacerbated the conduct of user studies due to public health restrictions. Therefore, this paper aims to leverage computer vision technologies to facilitate the readability evaluation in a VR environment.

Driven by deep neural networks and large-scale datasets, computer vision-based text detection and analysis methods have progressed substantially over the past years and continually improved their performances in multiple areas [9–11]. For example, Long et al. [11] used computer vision technologies and text-to-speech technologies on an Android smartphone to automatically identify and recognize texts and signs for visually impaired people. Computer vision has also been widely applied in the field of autonomous driving to replace human drivers to

perform tasks of recognition, motion estimation, tracking, and scene understanding [10,11]. The computer vision applications demonstrate that the gap between computer vision and human vision is getting smaller, and computer vision-based methods have the potential to facilitate the readability evaluation in VR.

2 LITERATURE REVIEW

Readability using electronic displays has been widely discussed since the 1980s. However, research on readability in VR through an HMD has been very limited [12] due to the recency of the commercialization of HMD technologies [13–15]. Grout et al. [16] investigated the possibility of using VR HMDs (Oculus Rift DK1 and Oculus Rift DK2) to replace a traditional monitor for daily computing tasks through a user study. The results showed that participants preferred a traditional monitor due to the familiarity and higher resolutions. This study indicated that rendering texts onto large flat panels significantly impaired reading performance in the VR headsets. Unsurprisingly, the results showed that participants with the use of the Oculus DK2 (960×1080 pixels/eye) performed better than those using the Oculus DK1 (640×800 pixels/eye).

Dingler et al. [1] conducted a user study with 18 participants to explore the impacts of text parameters (i.e., text size, convergence, view box dimension, and positioning) on the readability in VR. The study adopted the unit of distance-independent millimeter (dmm) [17] as a measure of text size. The results indicated that the recommended minimum height of characters should be 23 ± 7.8 dmm, and body height should be 41 ± 14 dmm. The study provided a set of recommendations as a reference point for other application developers. However, it focused on the UIs and the experiments were not designed for evaluating non-UI text on objects in the virtual environment.

Solum [3] conducted another user study with 24 participants to investigate how human, software, and hardware factors influence the readability in VR through an HMD. The results showed that the projection type of text in VR (flat/curved) had no significant impact on the character size for comfortable reading. But increased display qualities (resolution and field of view) can reduce the character size for reading. The experiment also indicated that the HMD required larger character sizes to achieve the same readability using standard computer monitors.

The readability issue of text display has been clearly identified through previous user studies and VR stakeholders are working on both software and hardware factors to improve text readability and provide a comfortable experience for users. For example, VR HMD manufacturers (e.g., Oculus, HTC, and Varjo) are improving hardware factors (e.g., screen size, resolution, and field of view) [4,18]; simulation platforms (i.e., Unity and Unreal) are optimizing the rendering pipelines and shaders [3,19]; VR content creators are exploring other settings, like text font type, size, color, projection type, placements, light conditions, etc. [5,20]. Due to the large (and evolving) number of factors at play, evaluation time and frequency required for user studies will increase exponentially, and user studies will become less efficient. In this study, we developed a

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new computer vision-based approach to evaluate text readability and compared its results with the findings from a user study.

3 METHODS

3.1 Experiment

In this study, we conducted a readability study on the impacts of texture mapping techniques on text-based traffic signs and compared the results of the cloud computer vision-based and human vision-based evaluation methods. The rationale for selecting text-based traffic signs is that signs are highly critical in developing situational awareness in transportation-related applications (e.g., a “DO NOT PASS” sign for an unseen hazard ahead). Besides, traffic signs follow strict guidelines for size, font, color, and placement (i.e., the Manual on Uniform Traffic Control Devices in the US) [21] and hold other readability factors constant.

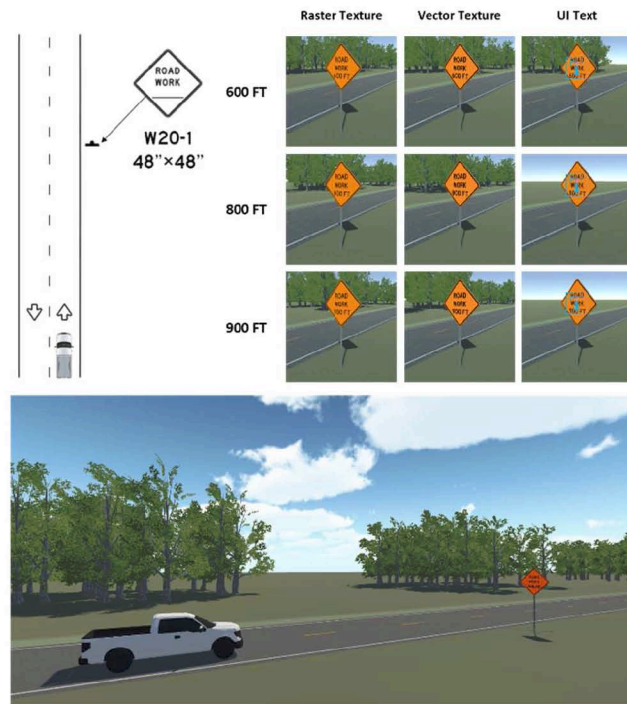


Figure 1: Experiment scenario and text-based traffic signs

The experiment was developed in Unity 3D (Version 2019.4.30f1) as a drive-through scenario (Figure 1). There were nine traffic work zone signs to drive past: three work signs (W20-1) with the text of ROAD WORK 600/800/900 FT for each texture mapping technique (raster texture/vector texture/UI text). A VR camera was set in front of a pickup truck, which was programmed to drive at a constant speed of 3 m/s toward traffic signs. All other environmental settings, including lights, cameras, and shaders,

were unchanged during this study. The experiments can evaluate if different texture mapping techniques have significant impacts on the readability of text-based signs in the VR environment.

3.2 Evaluation Methods

In order to examine the feasibility of using cloud-based computer vision service for readability evaluation, we used two evaluation methods for the same experiments and the framework of the two evaluation methods is present in Figure 2.

3.2.1 User study/Human vision

The study research protocols were evaluated and approved by the university institutional review board (IRB) in August 2021. Ten human participants were recruited through flyers and social networks, and the experiments were conducted in September 2021.

Apparatus Previous studies found that higher VR HMD resolution improved the quality of visual perception, immersion, and text readability [4,23,24]. We used the Oculus Rift S (1280 × 1440 pixels/eye) for all participants in the study. The computers used to run the experiments had an Intel i7-9750H processor, 16 GB of RAM, an Nvidia GeForce 2060 graphics card, and Windows 10 operating system.

Procedure After participants signed the consent form, they were asked to sit on a chair and wear the Oculus Rift S headset. As a passenger, the participant did not need to drive the vehicle to pass the nine text-based traffic signs. The appearance order of the nine traffic signs was randomly generated, so the participant did not predict the distance number on the sign (600/800/900 FT) based on his/her experiment experience, and it also helps avoid any possible sequence bias. Participants had the ability to pause for a break and stop the experiments at any time.

Measure Farthest reading distance was used to measure the readability of text-based traffic signs. Participants were asked to immediately report when they could see and read the distance information on the sign. The research team recorded the distance to the sign by pressing the hotkey after hearing participants' reports.

Effort Analysis When preparing for a user study, significant time and effort was spent on preparing research protocol and obtaining the IRB approval. For this study, the research protocol was approved within two weeks. However, a survey showed that the time from submission of an expedited IRB application to approval averaged 54.8 days in 68 hospitals in the US [25]. Besides, a considerable amount of time was spent on recruiting suitable participants and scheduling appointments with them. It took the research team six weeks to complete the experiments.

3.2.2 Cloud computer vision

Compared with an open-source library (e.g., OpenCV and PyTorch CV) for local computer vision analysis, cloud computer vision services (e.g., AWS, Google Cloud, and Microsoft Azure) could significantly reduce the time and effort required and provide more accurate results with the use of pre-trained and industry-leading technologies [11]. Therefore, in this study, we used the Google Cloud Vision API service (Figure 3) as an example to illustrate the application of the cloud computer vision-based approach.

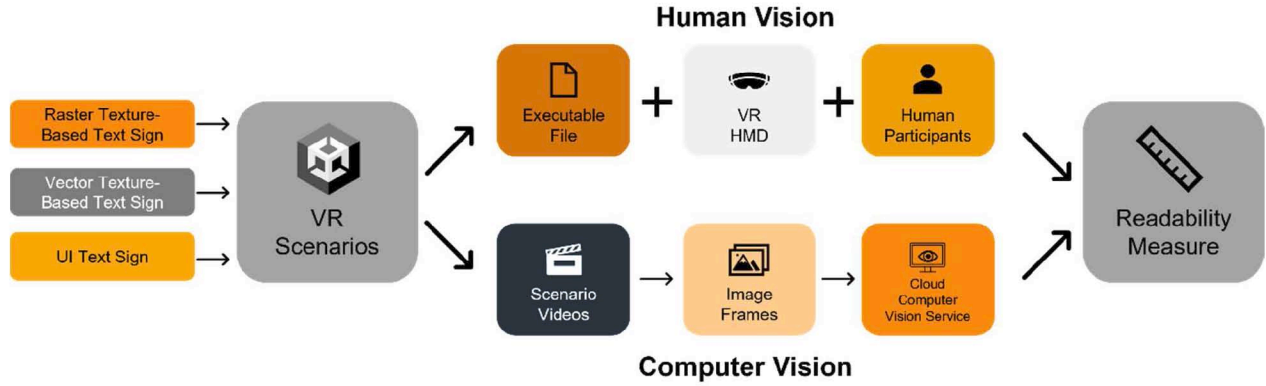


Figure 2: Human vision and computer vision evaluation

Procedure The workflow of the evaluation method is shown in Figure 2. The drive-through video was generated from the same scenario which was used for the user study. When the scenario video (30 frames/second) was exported, the resolution was set as the display quality of Oculus Rift S (1280×1440 pixels/eye). Then, the videos were saved as JPEG images frame by frame in chronological order. The generated JPEG files were uploaded to the Google Cloud Storage bucket and analyzed by the Cloud Vision service [26].

Measure Same as the user study, the farthest reading distance was measured. The results were automatically generated from the cloud platform and saved as JSON files. The reading distance was identified by searching the first frame with correct sign text.

Effort Analysis The biggest advantage of using the computer vision-based method is its efficiency. By following the Google Cloud step-by-step guides, the results were obtained within one day. However, using the Google Cloud Vision API service would incur certain fees. For example, we generated 1,200 images ($40 \text{ seconds video} \times 30 \text{ frames/second}$) for each sign, and a total of 10,800 images ($1,200 \text{ images/sign} \times 9 \text{ signs}$) were uploaded and analyzed by the computer vision service. Based on the current pricing information [27], the total cost for the study would be \$29.4 (\$3.00 label and text detection fee per 1,000 images after the first free 1,000 images usage).

(28.09/m), both vector texture (30.143m) and UI text (28.484 m) signs had slightly better performance on the reading distance measure. However, based on paired t-test results, the improvements were not statistically significant at 95% confidence level.

Table 1: Results of the user study

	Raster Texture	Vector Texture	UI Text
n	30	30	30
Mean (m)	28.097	30.143	28.484
SD	3.879	3.935	4.819
Difference	Baseline	2.046	0.387
P-value	Baseline	0.1233	0.6619

The results of the computer vision-based method are shown in Table 2. Again, both vector texture (28.217m) and UI text (29.943m) signs performed slightly better than the raster texture-based signs (27.843m). However, the p-values show that the differences between the different methods were not statistically significant (at 95% confidence level).

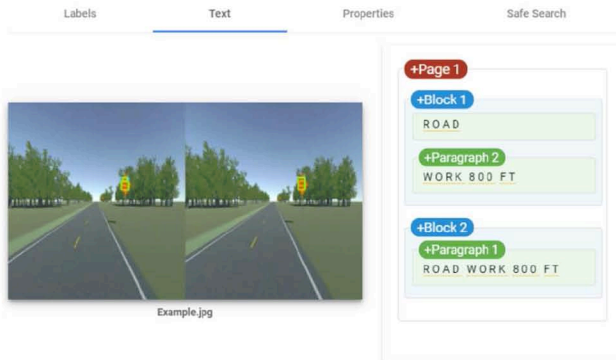


Figure 3: Example of Google Cloud Vision API

4 RESULTS

The user study results are shown in Table 1. As raster texture-based models are the most commonly used model type in Unity or any real-time rendering software [2, 28, 29], it was considered as the baseline model. Compared with the baseline raster texture

Table 2: Results of the computer vision-based method

	Raster Texture	Vector Texture	UI Text
n	3	3	3
Mean (m)	27.843	28.217	29.943
SD	0.200	1.274	1.026
Difference	Baseline	0.374	2.10
P-value	Baseline	0.6706	0.1381

The results shown in Tables 1 and 2 indicate the high similarities between the cloud computer vision and human vision methods. Both methods had similar farthest reading distances (i.e., the first time the text on a sign is readable) for all three texture models. In addition, the two methods found that both vector texture-based and UI text-based signs could slightly improve the farthest reading distance (0.387m - 2.046m for human vision, 0.374m - 2.10m for computer vision).

5 DISCUSSION

The readability experiments of different texture mapping techniques showed that the cloud computer vision-based evaluation method generated results that were consistent with the user study. While these results are promising, they come with a few caveats. First, text readability is subjective and it is highly related to the user's experiences and expectations [19,30]. For example, in a transportation-related VR application, users can recognize the "STOP" sign simply relying on the red color and its octagonal shape, which can in turn help them read the text without actually seeing the words. Second, the scenario videos could not include subtle head movements, which allow users to read texts from a comfortable angle. These two issues can be explored in future research.

The main purpose of developing the cloud computer vision-based evaluation method is to help VR developers and users quickly assess the impacts of various visual factors in VR. For example, in the study, we used a 2048×2048 pixels PNG file as the texture for the traffic sign. In order to reduce the size of the project, will it be acceptable to reduce the texture resolution to 512×512 pixels? The question is very practical, but it is not worth conducting a one-month-long user study to answer it. However, a computer vision-based evaluation method can be used to efficiently answer this question.

6 CONCLUSIONS

This paper presents the first results of a cloud computer vision-based readability evaluation method in VR. The performance of the new method was compared to a human user study on text-based traffic signs. The results showed that the performance of cloud computer vision was close to human vision, and the distances at which signs are readable were identical in both methods. By using the cloud computer vision service, developers are able to evaluate the impacts of various visual factors on text display without recruiting human participants and thus significantly reducing the overall effort and time to improve text readability experience in VR.

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